III SEMESTER	L	T	P	C
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20AM3T01 : DATA STRUCTURES AND ALGORITHMS				

COURSE OBJECTIVES:

1. To impart the basic concepts of data structures and algorithms.
2. To be familiar with writing recursive methods.
3. To implement operations on Linked List, Stack and Queues.
4. To implement traversal operations of trees and graphs.
5. To understand concepts about various algorithm design techniques, searching and sorting techniques.

COURSE OUTCOMES:

On completion of the course the student will be able to

CO1: Understand the concept of recursive algorithms.
CO2: Demonstrate the different types of data structures.
CO3: Able to understand the operations on linear data structures.
CO4: Summarize searching and sorting techniques.
CO5: Choose appropriate data structure as applied to specified problem definition.
CO6: Understand and implement the various algorithm design techniques.

UNIT 1:

INTRODUCTION TO ALGORITHMS: Introduction to Data vs Information - Data Structures - Classification – Abstraction - Abstract data types (ADT) - Array - characteristics - Storage Representations. Array Order Reversal – Recursion - Array operations, Algorithm - complexity – Time and Space trade off.

UNIT 2:

LINKED LIST: Array vs Linked List – Singly linked list - Representation of a linked list in memory - Operations on a singly linked list - Merging two singly linked lists into one list - Reversing a singly linked list – Polynomial Manipulation using List - Advantages and disadvantages of singly linked list - Circular linked list - Doubly linked list - Circular Doubly Linked List.

UNIT 3:

STACKS & QUEUES: Introduction – Array Representation of a Stack – Linked List Representation of a Stack - Stack Operations - Algorithm for Stack Operations - Stack Applications: Tower of Hanoi - Infix to postfix Transformation - Evaluating Arithmetic Expressions. Queue – Introduction – Array Representation of Queue – Linked List Representation of Queue - Queue Operations - Algorithm for Queue Operations -. Queue Applications: Priority Queue.

UNIT 4:

TREES AND GRAPHS: Preliminaries of Tree ADT - Binary Trees - The Search Tree ADT–Binary Search Trees - AVL Trees - Tree Traversals - B-Trees - Heap Tree – Preliminaries of Graph ADT - Representation of Graph – Graph Traversal - BFS – DFS – Applications of Graph – Shortest - Path Algorithms – Dijkstra's Algorithm, Minimum Spanning Tree – Prims Algorithm

UNIT 5:

ALGORITHM DESIGN TECHNIQUES & SEARCHING AND SORTING TECHNIQUES: Divide and Conquer Strategy – Greedy Algorithm – Dynamic Programming – Backtracking Strategy - List Searches using Linear Search - Binary Search - Fibonacci Search - Sorting Techniques - Insertion sort - Heap sort - Bubble sort - Quick sort - Merge sort - Analysis of sorting techniques.

TEXT BOOKS:

1. Jean-Paul Tremblay, Paul G. Sorenson, 'An Introduction to Data Structures with Application', TMH, 2017.
2. Richard F, Gilberg, Forouzan, "Data Structures", Cengage, 2nd Edition, 2004.

REFERENCE BOOKS:

1. Larry R. Nyhoff, ADTs, Data Structures, and Problem Solving with C++, Prentice Hall Edition, 2004.
2. Thomas H. Cormen, Charles E. Leiserson, "Introduction to Algorithms", 3rd Edition, 2010.